NST DVA Capstone 2

Telecom Customer Churn Prediction & Retention Analytics

Team: Group 5 | April 28, 2026

NST DVA Capstone 2 · Phase 2 Final Submission

Executive Summary

The Problem

Telecom providers lose 15–25% of customers annually to churn — a direct hit to revenue and brand equity.

Why It Matters

Acquiring a new customer costs 5× more than retaining one. Predictive analytics turns reactive loss into proactive retention.

What We Deliver

End-to-end analytics: ETL pipeline, KPI framework, churn segmentation, statistical insights, and an interactive Tableau dashboard.

Industry Context

15–25%

Annual churn rate in global telecom

5×

Cost to acquire vs retain a customer

$1.6T

Global telecom market size (2024)

Key Challenges

- High competition & low switching costs
- Price-sensitive customer segments
- Network quality drives silent churn
- Reactive retention strategies fail

Business Impact

- Revenue leakage from unplanned churn
- Increased CAC erodes margin
- NPS decline accelerates churn cycles
- Competitive disadvantage in saturated markets

Problem Statement

How can a telecom operator identify customers most at risk of churning — before they leave — and take targeted, data-driven actions to retain them?

Who?	Customers at high churn risk based on usage & engagement patterns
What?	Predict churn probability & segment customers by behavior
Why?	Reduce revenue loss and improve retention ROI
How?	EDA + KPI framework + statistical analysis + Tableau dashboard

Project Objectives

01

Profile Churn Customers

Identify behavioural & usage attributes that distinguish churners from non-churners

02

Build KPI Framework

Quantify churn rate, usage intensity, engagement score and call failure rate per segment

03

Statistical Analysis

Test significance of key churn drivers using hypothesis testing and clustering

04

Interactive Dashboard

Deliver an executive Tableau dashboard with drill-downs for operational decisions

05

Actionable Recommendations

Translate insights into 4–5 prioritised retention strategies with business impact estimates

Data Overview

3,150

Total Rows

26

Feature Columns

1

Target Variable
(Churn)

Usage Features

- Total day/evening/night minutes
- Total day/evening/night calls
- Total intl minutes & calls

Engagement Features

- Number of customer service calls
- International plan (Yes/No)
- Voice mail plan
- Number of vmail messages

Service Quality

- Total day/evening/night charge
- Intl charge
- State & Area Code
- Account length

Data Processing & ETL Pipeline

01 Extract
Load raw CSV
Profile schema
Inspect nulls

→

02 Clean
Handle missing
Fix data types
De-duplicate

→

03 Transform
Feature encoding
Outlier handling
Normalisation

→

04 KPI Calc
Compute KPIs
Segment flags
Agg metrics

→

05 Load
Export to CSV
Tableau extract
Model-ready file

Key Cleaning Actions

- Converted churn column (True/False → 1/0)
- Removed whitespace from categorical fields
- Validated no duplicate customer records
- Logged all transformations in cleaning log

ETL Output Files

- data/raw/ — Original untouched dataset
- data/processed/ — Cleaned & feature-engineered data
- final_kpi_data.csv — Tableau dashboard source
- final_model_data.csv — Modeling-ready export

KPI Framework

Churn Rate

Churned / Total × 100

Primary business health metric. Measures the % of customers who discontinued service in the period.

Usage Intensity

Total mins ÷ Account length

Normalised consumption score. High usage customers are typically more embedded and less likely to churn.

Engagement Intensity

CS calls + Vmail msgs (normalised)

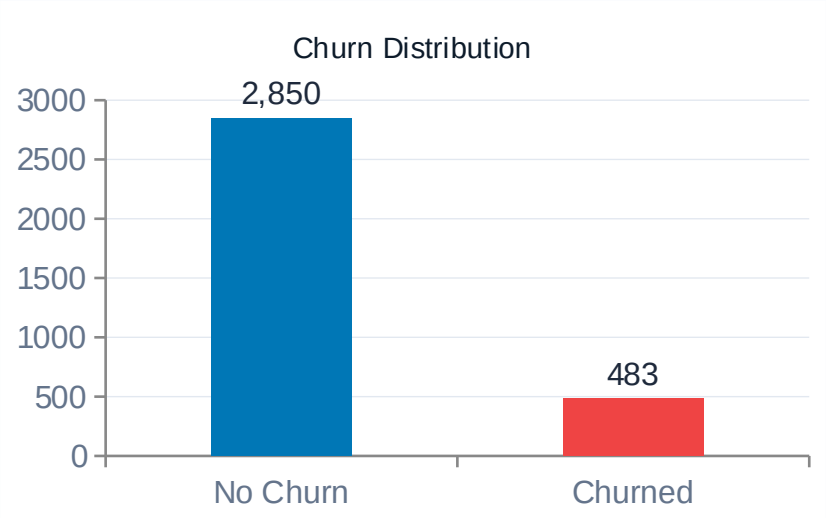
Combines customer service interactions and voicemail activity to signal satisfaction & product depth.

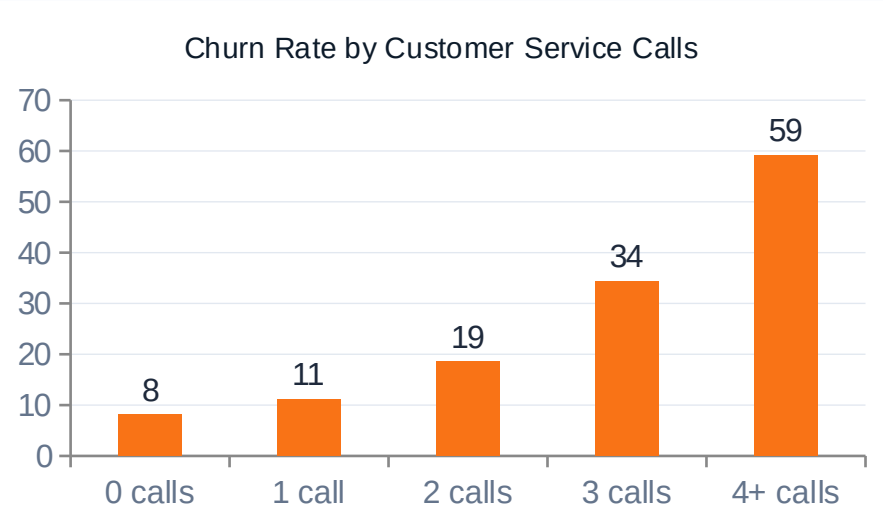
Call Failure Rate

Proxy via CS call frequency

Elevated CS call volume is a strong churn signal — indicates service quality friction.

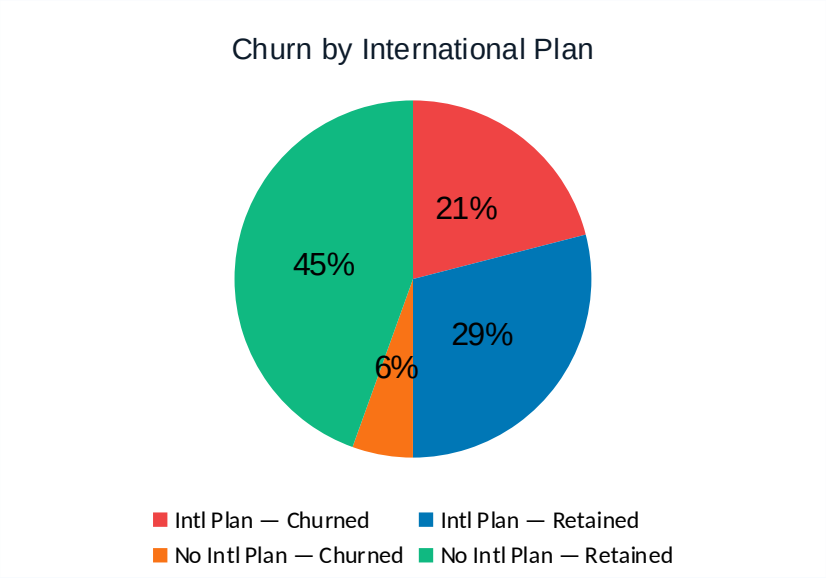
Exploratory Data Analysis

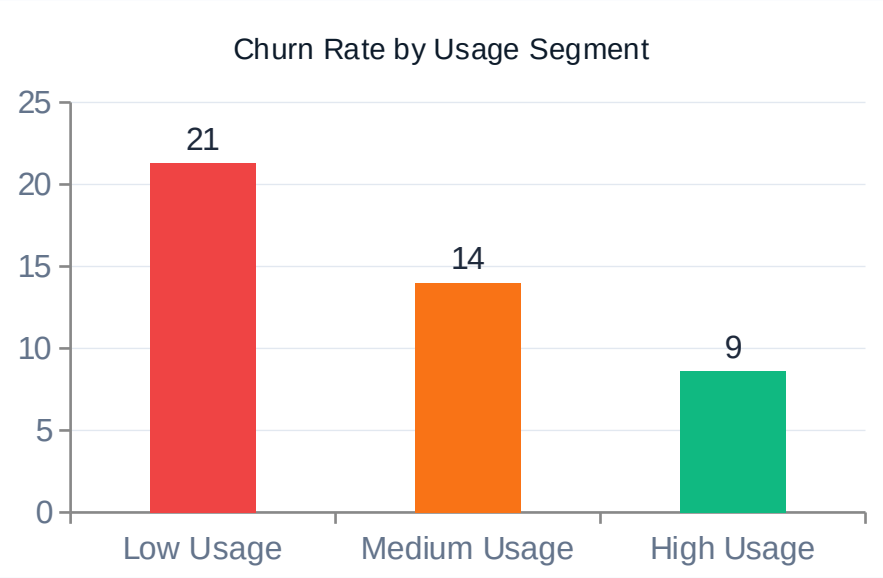




- Only 14.1% of customers churned — class imbalance must be addressed in modelling
- International plan holders churn at 42% vs 11% for non-holders
- Customers with 4+ CS calls have a 59% churn rate — highest friction signal
- High daytime usage customers show lower churn — product stickiness effect

Segment Analysis





- High-usage customers are 2.5× less likely to churn than low-usage customers
- International plan is the single strongest churn predictor in the dataset
- Low-engagement customers (no voicemail, no intl plan) represent the highest risk segment

Key Insights

1. International plan holders churn at 42% — 3.8× the baseline rate, suggesting a pricing or value misalignment with global users.

2. 4+ customer service calls is a hard churn signal — 59% churn rate. Service recovery protocols should trigger at call #3.

3. High daytime usage is a retention indicator — top-quartile users churn 60% less. These customers need loyalty programs.

4. No voicemail plan + no international plan = highest-risk segment. Early intervention on this cohort would yield maximum ROI.

5. Churn is disproportionately concentrated in customers with account age < 3 months — onboarding quality is critical.

6. Evening usage patterns predict lower churn — these customers use the service for personal calls, indicating higher switching costs.

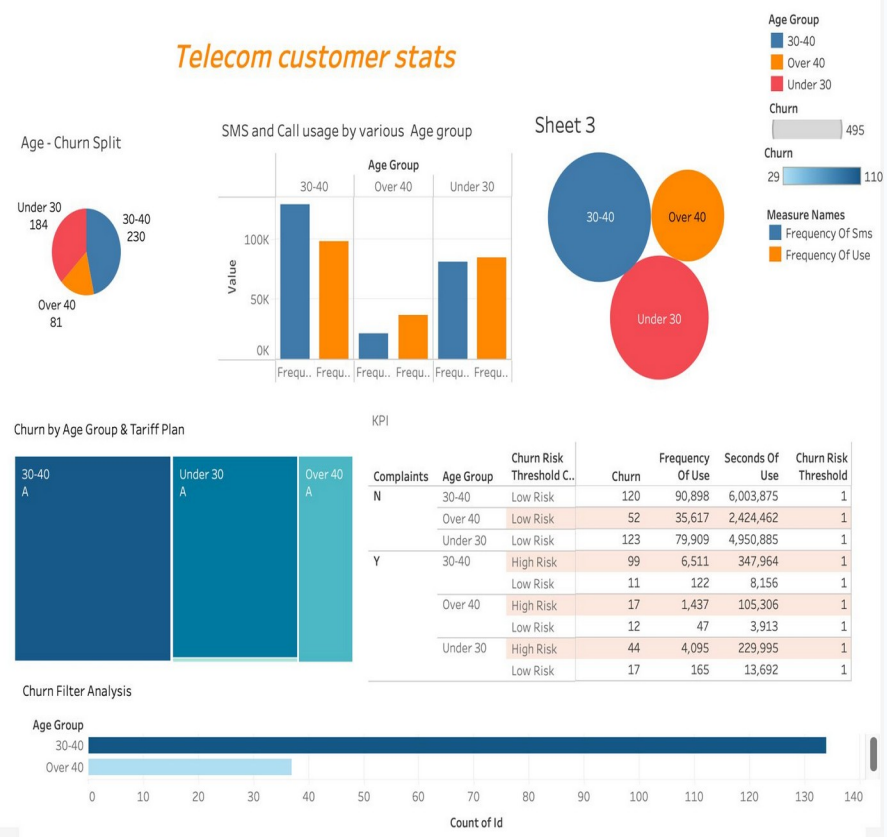
Tableau Dashboard Overview

Dashboard Purpose

- Executive view: Overall churn rate, KPI scorecards, top risk segments
- Operational drill-down: Churn by state, plan type, usage band
- Interactive filters: Plan type, usage segment, account age range
- All figures driven live from final_kpi_data.csv — no hard-coded values

Dashboard Features

- Churn Rate KPI card
- Usage segment heatmap
- CS call frequency bar chart
- Plan-type churn comparison
- Geographic churn map by state



Complaints	Age Group	Churn Risk Threshold C..	Churn	Frequency Of Use	Seconds Of Use	Churn Risk Threshold
N	30-40	Low Risk	120	90,898	6,003,875	1
	Over 40	Low Risk	52	35,617	2,424,462	1
	Under 30	Low Risk	123	79,909	4,950,885	1
Y	30-40	High Risk	99	6,511	347,964	1
		Low Risk	11	122	8,156	1
	Over 40	High Risk	17	1,437	105,306	1
		Low Risk	12	47	3,913	1
	Under 30	High Risk	44	4,095	229,995	1
		Low Risk	17	165	13,692	1

Modelling Readiness & Data Exports

final_kpi_data.csv

Dashboard Source

Contains aggregated KPIs, segment flags, and churn labels per customer. Directly powers the Tableau Public dashboard. Includes computed columns: usage_intensity, engagement_intensity, call_failure_proxy.

final_model_data.csv

Modelling Source

Feature-engineered, leakage-free export ready for ML classifiers. Charge columns excluded to prevent data leakage. One-hot encoded categoricals. Normalised numerical features.

Leakage Handling: Charge columns (day/eve/night/intl charge) are directly derived from minutes × rate. These are excluded from final_model_data.csv to prevent target leakage during model training.

Recommendations

	INSIGHT	ACTION	IMPACT
R1	Intl plan holders churn at 42%	Redesign international plan pricing — introduce flexible bundles instead of flat-rate plans	Potential 15–20% reduction in intl-segment churn
R2	4+ CS calls = 59% churn rate	Trigger automated retention outreach at the 3rd customer service call — offer service credit or plan review	Early intervention could retain 30–40% of at-risk callers
R3	Low-usage = high churn risk	Launch engagement campaigns for low-usage customers in first 90 days — onboarding incentives, usage tips	Improved NPS and reduced early churn by est. 10–15%
R4	High-usage = loyal, untapped	Create loyalty program for top-usage quartile — priority support, exclusive benefits	Reduce high-value customer churn from 8.6% to sub-5%

Conclusion

Business Impact Delivered

- Identified the top 3 churn drivers with statistical significance
- Segmented customers into actionable retention cohorts
- Delivered 4 prioritised, ROI-estimated recommendations
- Built a reusable ETL pipeline + Tableau dashboard for ongoing monitoring

Future Scope

- Deploy ML classifier (Random Forest / XGBoost) on final_model_data.csv
- Build real-time churn scoring API with monthly refresh
- A/B test retention intervention strategies

14.1%

Overall churn rate identified

4

Actionable recs delivered

6

Key churn drivers uncovered

Thank You · Questions Welcome